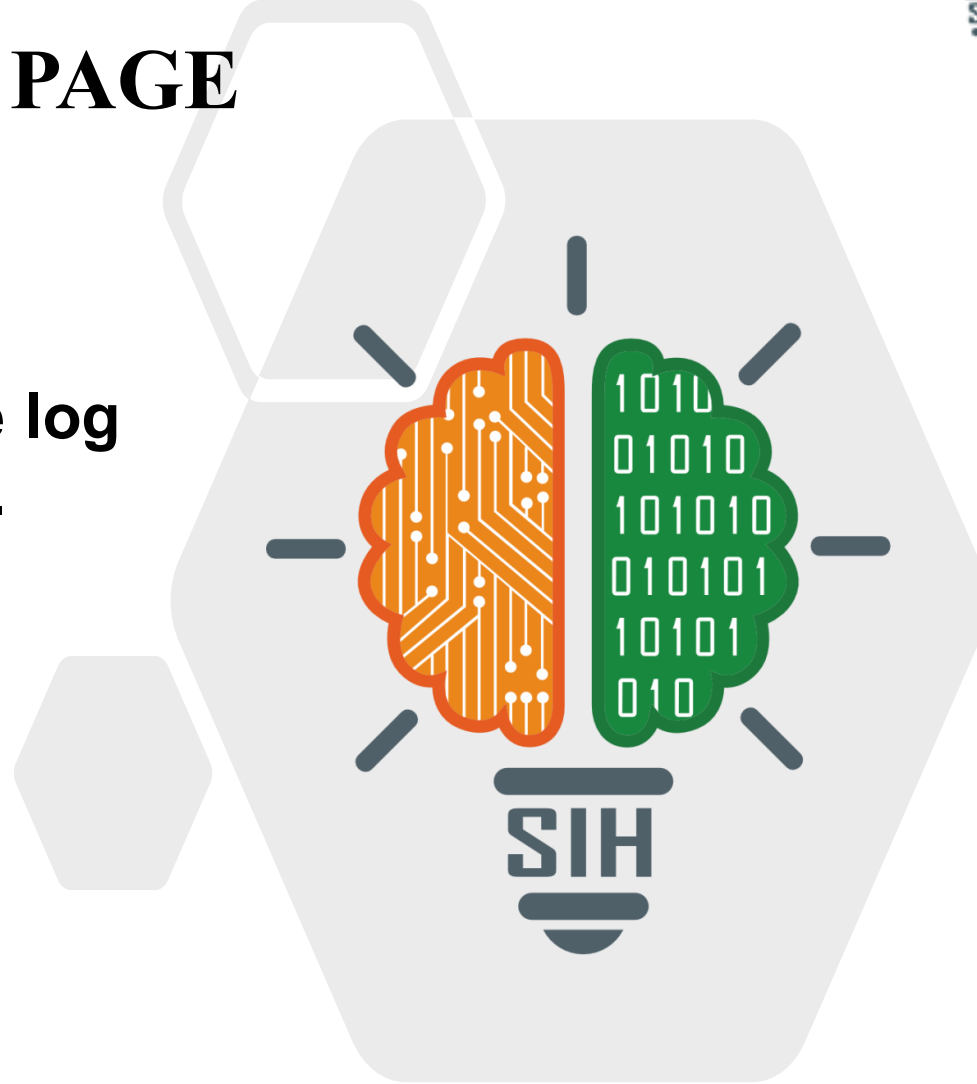


SMART INDIA HACKATHON 2025



TITLE PAGE

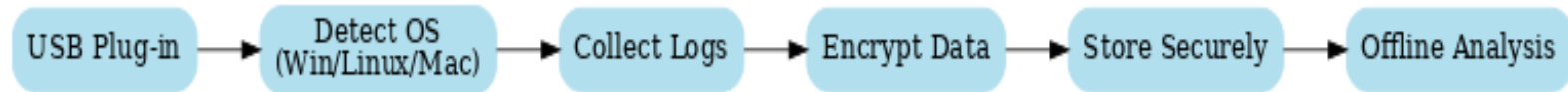
- **Problem Statement ID- SIH25235**
- **Problem Statement Title- Portable log analysis tool for isolated network.**
- **Theme- Smart Automation**
- **PS Category- Software**
- **Team ID- 53**
- **Team Name- Phoenix**



❖ Proposed Solution

Overview

- A **portable USB-based tool** that **auto-detects the operating system** (Windows/Linux/Mac).
- **Collects system and network logs, encrypts them, and stores securely** for offline analysis.



Problem Solving

- Works **seamlessly in air-gapped or isolated networks** (no internet needed).
- Ensures **secure, portable, and uniform log monitoring** across systems.
- Enables **centralized review** while maintaining **data confidentiality**.

Innovation

- **Rubber-duddy style automation** for **OS detection and log collection**.
- Uses a **proprietary encrypted format**, accessible only via its **companion software**.
- Acts as a **“Portable SOC-in-a-USB”**, combining **collection, security, and offline analysis** in one solution.

Programming Languages

- **Python** → Cross-platform scripting
- **Bash / PowerShell** → OS-specific automation
- **C / C++** → Performance-critical modules

Frameworks & Libraries

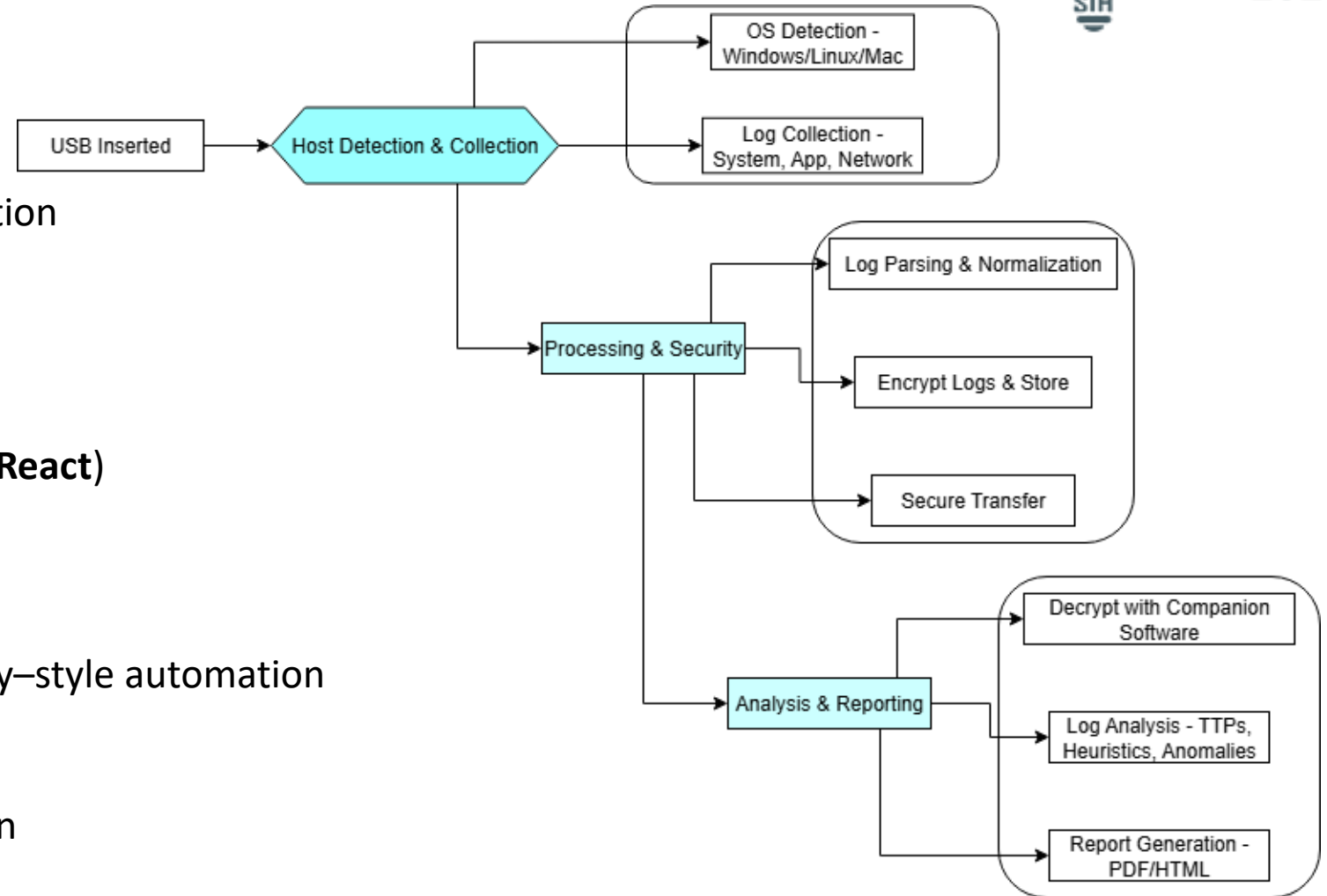
- **Log Parsing** → Regex, **Pandas**, PyParsing
- **Encryption** → **PyCryptodome**, OpenSSL
- **GUI** → Tkinter, **PyQt**, Web-based (**Flask / React**)

Hardware

- **USB drive** with autorun capability
- **Portable storage** ≥ 16GB
- **Optional microcontroller** → Rubber Ducky-style automation

Other Tools

- **Syslog / FTP collectors** for log transfer
- **YARA rules** → anomaly/malware detection
- **SQLite** → lightweight local log storage



FEASIBILITY AND VIABILITY



1. Feasibility

- **Technical** → Python, Bash/PowerShell + encryption libraries; **easy USB deployment**
- **Operational** → Works **offline in air-gapped networks**
- **Financial** → **Low-cost** (USB + open-source software)
- **Scalability** → **Easily replicable** across networks

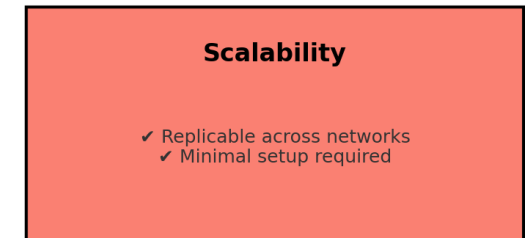
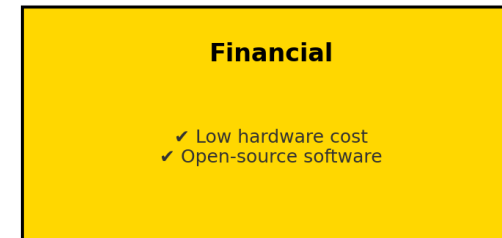
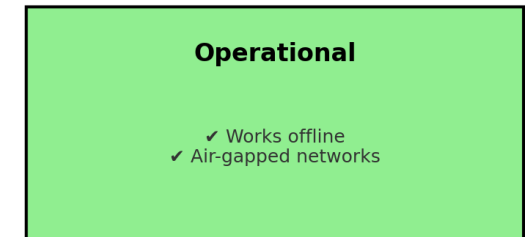
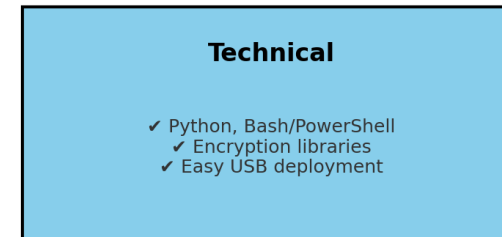
Feasibility & Viability

2. Challenges & Risks

- **OS Compatibility** → Scripts may behave differently
- **Security Risks** → USB misuse / malware spread
- **Encryption Issues** → Lost keys → inaccessible logs
- **Hardware Limits** → Storage & speed constraints
- **User Errors** → Incomplete collection by non-tech staff

3. Mitigation Strategies

- **Cross-platform testing** & modular scripts
- **Secure USB handling** (read-only, signatures, malware scan)
- **Strong encryption & key management** with backups
- **Optimized storage** → compression & selective logs
- **User-friendly GUI** → guidance & alerts

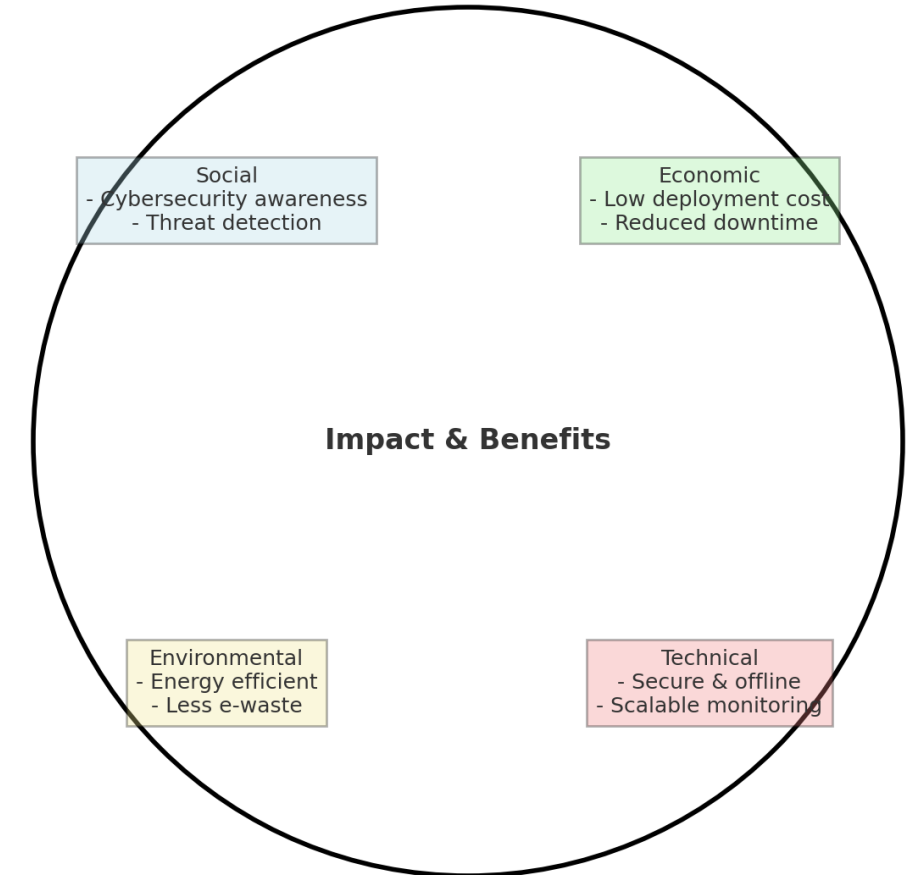


1. Target Audience Impact

- **IT & Security Teams:** Simplifies monitoring in isolated networks
- **Sensitive Organizations:** Secure log collection without leakage
- **SMEs:** Low-cost, portable SOC functionality
- **Auditors/Compliance:** Easier verification & reporting

2. Benefits

- **Social:** Improves awareness, reduces cyber-attack risks
- **Economic:** Lowers SOC costs, reduces downtime/losses
- **Environmental:** Energy-efficient, less e-waste via USB reuse
- **Technical:** Secure, offline, scalable; standardized log analysis



1. USB Artifact Analysis (Windows Event Viewer)

- Forensic analysis of USB devices and log collection

 <https://www.mdpi.com/2079-9292/8/11/1322>

2. Insider Threat Detection (Log Analysis & Event Correlation)

- Detecting malicious insider activities through log analysis

 https://www.researchgate.net/publication/276893438_Insider_Threat_Detection_Using_Log_Analysis_and_Event_Correlation

3. Online–Offline Framework for Anomaly Scoring

- Framework for detecting network anomalies in offline mode

 https://www.researchgate.net/publication/348438948_An_Online_Offline_Framework_for_Anomaly_Scoring_and_Detecting_New_Traffic_in_Network_Streams

4. Survey on Automated Log Analysis (Reliability Engineering)

- Techniques for automated log analysis in systems

 <https://pinjiahe.github.io/files/pdf/research/CSUR21.pdf>

5. USB Storage Device Forensics (Windows 10)

- Methods to analyze USB device activity using logs & registry

 <https://onlinelibrary.wiley.com/doi/10.1111/1556-4029.13596>